

Sargassum & Household Welfare in the Dominican Republic

Regression Tables for Paper

How to use: Open this file in any web browser (Safari, Chrome, etc.). Click just before a table title, drag to select the whole table, then copy (⌘C on Mac) and paste directly into Word or Google Docs. The table formatting will be preserved. To select all tables at once: Edit → Select All, then paste.

Panel: 100 municipalities × 2017–2025 × monthly (10,263 observations).
All models include municipality fixed effects and year×month fixed effects, with standard errors clustered by municipality.

Table 1. Effect of Sargassum Exposure on Household Welfare

	(1)	(2)	(3)
	Log Income	Employment Rate	Log Income
Dependent variable:	log(income)	employment rate	log(income) — peak
Sargassum intensity	-28.148 (18.527)	-4.439 (3.937)	-73.896** (24.281)
Observations	10,263	10,263	4,344
R²	0.522	0.305	0.593

Notes: All models estimated by OLS with municipality and year×month fixed effects. Standard errors clustered by municipality in parentheses. Treatment variable is mean excess AFAI above the 0.001 threshold (Sargassum intensity). Peak season = May–September. Within R² measures variation explained after removing fixed effects. * p<0.05, ** p<0.01, *** p<0.001.

Table 2. Sargassum Effect by Income Tertile (Full Year and Peak Season)

	(1)	(2)	(3)	(4)	(5)	(6)
	Full Year			Peak Season (May–Sep)		
	T1 (Bottom)	T2 (Middle)	T3 (Top)	T1 (Bottom)	T2 (Middle)	T3 (Top)
Dependent variable:	log income T1	log income T2	log income T3	log income T1	log income T2	log income T3
Sargassum intensity	-144.866 (90.053)	-30.749* (12.709)	-12.198 (21.736)	-75.071 (147.628)	-48.995** (16.472)	-26.640 (31.002)
Observations	10,122	10,091	9,777	4,287	4,266	4,154
R²	0.185	0.457	0.421	0.214	0.490	0.446

Notes: All models estimated by OLS with municipality and year×month fixed effects. Standard errors clustered by municipality in parentheses. Tertiles defined nationally by year using per-capita household income. T1 = bottom third, T2 = middle third, T3 = top third. Dependent variable is log mean income of each tertile group within the municipality–month cell. * p<0.05, ** p<0.01, *** p<0.001.

Table 3. Sargassum Effect by Coastal Proximity (Full Year and Peak Season)

	(1)	(2)	(3)	(4)
	Full Year		Peak Season (May–Sep)	
	Coastal	Not Coastal	Coastal	Not Coastal
Dependent variable:	log income	log income	log income	log income
Sargassum intensity	-28.841 (49.173)	-22.801 (17.154)	-58.947 (61.190)	-80.076** (25.533)
Observations	4,767	5,496	2,025	2,319
R²	0.462	0.565	0.600	0.580

Notes: All models estimated by OLS with municipality and year×month fixed effects. Standard errors clustered by municipality in parentheses. Coastal municipalities have direct ocean shoreline contact (48 municipalities); not-coastal municipalities are all others (52 municipalities). The Haiti land border is excluded from the coastline classification. * p<0.05, ** p<0.01, *** p<0.001.

Column headers for Tables 2 and 4: Columns (1)–(3) report full-year results; columns (4)–(6) report peak-season (May–Sep) results.

Column headers for Table 3: Columns (1)–(2) report full-year results; columns (3)–(4) report peak-season results.

Table 4. Sargassum Effect by Income Tertile — Positive-Income Households Only

	(1)	(2)	(3)	(4)	(5)	(6)
	Full Year			Peak Season (May–Sep)		
	T1 (Bottom)	T2 (Middle)	T3 (Top)	T1 (Bottom)	T2 (Middle)	T3 (Top)
Dependent variable:	log income T1	log income T2	log income T3	log income T1	log income T2	log income T3
Sargassum intensity	-14.173 (14.108)	-17.829 (16.849)	-7.296 (24.912)	-48.686** (18.029)	-27.019 (19.618)	-17.691 (35.587)
Observations	10,046	9,893	9,556	4,248	4,187	4,052
R²	0.341	0.452	0.408	0.365	0.461	0.433

Notes: All models estimated by OLS with municipality and year×month fixed effects. Standard errors clustered by municipality in parentheses. Tertiles defined nationally by year after **excluding zero-income households** (~17.6% of the sample). T1 = bottom third, T2 = middle third, T3 = top third of positive-income households. * p<0.05, ** p<0.01, *** p<0.001.

Appendix

Table A1. National Income Tertile Cutoffs by Year

	All households		Positive-income households only	
Year	T1/T2 boundary	T2/T3 boundary	T1/T2 boundary	T2/T3 boundary
2017	3,000	7,000	4,000	8,000
2018	3,225	7,740	4,300	8,859
2019	3,500	8,333	4,750	9,478
2020	2,867	7,500	4,300	8,750
2021	3,413	8,150	5,000	9,667
2022	3,933	9,213	5,375	10,500
2023	4,300	10,750	6,323	12,500
2024	5,000	12,000	7,000	14,000
2025	5,200	13,100	7,500	15,050

Notes: Units are RD\$ per capita per month (household total income divided by number of members). Cutoffs are national percentiles of the per-capita income distribution within each year. Positive-income sample excludes the approximately 17.6% of surveyed households reporting zero income.